

Critical-Window and Terminal-Prime Contraction for the Alaoglu–Erdős Conjecture

A nonduplicative progress report beyond the unified report

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Abstract

Assume hypothetically that $x \in \mathbb{R} \setminus \mathbb{Z}$ and $M = 2^x$, $A = 3^x$ are integers. The attached unified report introduced the falling-factorial cocycles

$$F_n = (A^n)_{M^n} = \frac{A^n!}{(A^n - M^n)!}, \quad \mathcal{F}_{P,n} = \prod_j F_{n+j}^{c_j}, \quad P(T) = \sum_j c_j T^j,$$

and reduced their archimedean near-unit property to the annihilations $P(M) = P'(M) = P(M^2/A) = 0$. It left a linear-width feasibility problem: could a varying annihilator $P = P_n$ make $\mathcal{F}_{P,n}$ both integral and asymptotic to 1 after crossing the prime-window span threshold?

This report closes that feasibility problem negatively. The key is not a sharper estimate for one highest negative coefficient, but a simultaneous sum over the terminal-prime window attached to *every* negative coefficient. Integrality forces each negative M -weighted coefficient mass to be transported upward. The transport operator has norm at most

$$\kappa_n = \frac{2(M/A)^n}{1 - M/A} + \frac{2}{M-1} + \frac{2M}{n(M-1)^2} = o(1) + \frac{2}{M-1},$$

whereas $P(M) = 0$ says that the total positive and negative M -weighted masses are exactly equal. Since the independently verified finite region gives $M \geq 2^{14}$, one has $\kappa_n < 1$; hence no nonzero integer polynomial with $P(M) = 0$ can produce an integral falling-factorial cocycle for all sufficiently large n . The argument is uniform in the degree, span, coefficient height, and number of sign changes. It also applies to binomial cocycles.

Two supporting results are developed. First, the earlier prime-window theorem is sharpened from a fixed linear margin to any additive margin tending to infinity. Second, confluent Hermite interpolation identifies the canonical maximal-contact stencils and proves an exact “terminal-prime multiplicity mismatch”: a double root at M forces a boundary coefficient at least $(D+1)M^D$, while a transported terminal interval supplies only $O((M/A)^t M^D)$ copies of each terminal prime. These results explain why the apparent critical Hermite families cannot escape. They do not prove the Alaoglu–Erdős conjecture; they remove one of the unified report’s four explicit open directions and leave a more focused search for a genuinely primitive-sensitive mechanism.

New result — headline

The main theorem, [Theorem 4.1](#), gives a degree-free negative solution of the unified report’s “linear-width factorial feasibility” target. It uses only the imported candidate identities, Huxley’s theorem on primes in short intervals, elementary valuation bounds, and the exact mass balance $P(M) = 0$.

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1 Scope, audit boundary, and status

This document is deliberately not another survey of the conjecture. The attached unified report is more than twenty-three thousand lines long and already treats the standard transcendence, Kummer, determinant, interpolation, spectral, dynamical, p -adic, and formalization routes. Repeating those sections would conceal rather than clarify new progress. We import only the small interface needed below.

Imported prerequisite

Under failure of the conjecture, choose the least nonintegral solution (or simply fix any hypothetical nonintegral solution) and put

$$M = 2^x \in \mathbb{Z}_{>0}, \quad A = 3^x \in \mathbb{Z}_{>0}.$$

Then

$$\frac{\log M}{\log A} = \frac{\log 2}{\log 3} =: \alpha, \quad \frac{1}{2} < \alpha < \frac{2}{3}, \quad 1 < M < A.$$

The kernel-checked finite region in the attached report gives $x \geq 14$, hence $M \geq 2^{14}$ and

$$q := \frac{M}{A} = \left(\frac{2}{3}\right)^x \leq \left(\frac{2}{3}\right)^{14} = 0.0034254873 \dots$$

The report also proved the fixed-polynomial near-unit classification for the factorial cocycles used here.

The following status terminology is used.

Imported	proved or kernel-checked in the attached unified report; not reproved in full.
New theorem	proved in this report at the paper level.
Known analytic input	a classical external theorem, principally Huxley's short-interval prime theorem or Brun–Titchmarsh.
Open target	a precisely stated step not proved here.

The main new theorem is control-blind: it applies equally to the genuine integer-exponent pairs $(2^m, 3^m)$. That is appropriate because it is a *negative closure* of an auxiliary strategy, not a proof of the conjecture. A final proof must still use information that distinguishes a primitive hypothetical pair from the controls, such as an external support prime, least-generator structure, or the saturated rational-output spectrum.

2 Factorial cocycles and the first decay modes

Set

$$N_n = A^n, \quad K_n = M^n, \quad F_n = (N_n)_{K_n} = N_n(N_n - 1) \cdots (N_n - K_n + 1).$$

For a nonzero polynomial $P(T) = \sum_{j=0}^d c_j T^j \in \mathbb{Z}[T]$ define

$$\mathcal{F}_{P,n} = \prod_{j=0}^d F_{n+j}^{c_j}.$$

The expression is a positive rational number, and

$$v_p(\mathcal{F}_{P,n}) = \sum_j c_j V_p(n+j), \quad V_p(t) := v_p(F_t).$$

The mode notation used below is

$$\lambda = \frac{M^2}{A} > 1, \quad \eta = \frac{M^3}{A^2} < 1, \quad q = \frac{M}{A} < 1, \quad \tau = \frac{M^4}{A^3} < q, \quad \sigma = \left(\frac{M}{A}\right)^2 < \tau, \quad v = \frac{M^5}{A^4} < \sigma.$$

Their order follows from the elementary base inequalities $4 > 3$, $8 < 9$, $2/3 > 16/27 > 4/9 > 32/81$.

Lemma 2.1 (Six-term factorial expansion). *As $n \rightarrow \infty$,*

$$\log F_n = nM^n \log A - \frac{1}{2}\lambda^n - \frac{1}{6}\eta^n + \frac{1}{2}q^n - \frac{1}{12}\tau^n + \frac{1}{4}\sigma^n + O(v^n). \quad (1)$$

The implied constant depends only on M, A .

Proof. Write $N = A^n$, $K = M^n$, and $u = K/N = q^n$. Then

$$\log F_n = K \log N + \sum_{j=0}^{K-1} \log \left(1 - \frac{j}{N}\right).$$

For $0 \leq y \leq u < 1/2$,

$$\log(1-y) = -y - \frac{y^2}{2} - \frac{y^3}{3} + O\left(\frac{y^4}{1-u}\right).$$

Using

$$\begin{aligned} \sum_{j < K} j &= \frac{K^2 - K}{2}, \\ \sum_{j < K} j^2 &= \frac{2K^3 - 3K^2 + K}{6}, \\ \sum_{j < K} j^3 &= \frac{K^4 - 2K^3 + K^2}{4}, \end{aligned}$$

produces the displayed modes. The Taylor remainder is $O(K^5/N^4) = O(v^n)$. The omitted lower-power terms from the three displayed power sums have bases at most $(8/27)^x < (32/81)^x$, so they are absorbed by the same error. $\square$

Let E denote the forward shift. The identities

$$P(E)(\rho^n) = P(\rho)\rho^n, \quad P(E)(n\rho^n) = \rho^n(nP(\rho) + \rho P'(\rho))$$

show why the imported near-unit classification begins with

$$P(M) = P'(M) = P(\lambda) = 0.$$

For a fixed P , these conditions remove the visible term and the unique divergent collision mode. The remaining leading mode is usually η^n . Varying $P = P_n$ can cancel further modes, but its coefficients and degree then grow, and uniformity becomes the entire issue.

3 Terminal-prime windows

For an index t put

$$X_t = A^t, \quad H_t = M^t = X_t^\alpha, \quad I_t = (X_t - H_t, X_t].$$

Huxley's theorem applies because

$$\alpha = \frac{\log 2}{\log 3} = 0.6309297535 \dots > \frac{7}{12}.$$

Consequently

$$\vartheta(X_t) - \vartheta(X_t - H_t) = (1 + o(1))H_t, \quad (2)$$

where $\vartheta(y) = \sum_{p \leq y} \log p$. In particular, I_t contains primes for every sufficiently large t .

A prime in I_t occurs in exactly one place in the base factorial F_t . Its possible reappearance in a higher factorial is controlled uniformly as follows.

Lemma 3.1 (Transported terminal-prime valuation). *There is $t_0 = t_0(M, A)$ such that for every $t \geq t_0$, every prime $p \in I_t$, and every integer $\ell \geq 1$,*

$$V_p(t + \ell) \leq 2q^t M^\ell + 2 \left(1 + \frac{\ell}{t}\right). \quad (3)$$

Moreover $V_p(t) = 1$, and $V_p(s) = 0$ for $0 \leq s < t$.

Proof. For large t one has $p > X_t - H_t > X_t/2$, so p and no other multiple of p lies in the terminal interval defining F_t , and $p^2 > X_t$. Also $A^{t-1} < A^t - M^t$, so no lower factorial contains p .

For $s = t + \ell$, put $Y = A^s$, $W = M^s$. Legendre's interval formula gives

$$V_p(s) = \sum_{r \geq 1} \left(\left\lfloor \frac{Y}{p^r} \right\rfloor - \left\lfloor \frac{Y - W}{p^r} \right\rfloor \right).$$

Each nonzero summand is at most $W/p^r + 1$. Hence

$$V_p(s) \leq \frac{W}{p-1} + \frac{\log Y}{\log p}.$$

Since $p \geq A^t(1 - q^t)$, the first term is at most $2q^t M^\ell$ for large t . The second is at most $2(1 + \ell/t)$, again uniformly for large t . $\square$

The first term in (3) is the expected number of transported multiples. The second is the crude one-per-prime-power allowance. After M -weighting and summing over coefficient indices, both kernels are summable. This observation is the decisive step.

4 The global terminal-prime contraction

For a coefficient c_j write $c_j^+ = \max(c_j, 0)$ and $c_j^- = \max(-c_j, 0)$. Define the positive and negative M -weighted masses

$$S_+(P) = \sum_j c_j^+ M^j, \quad S_-(P) = \sum_j c_j^- M^j.$$

If $P(M) = 0$, then

$$S_+(P) = S_-(P) > 0. \quad (4)$$

The strict positivity follows because a nonzero polynomial with positive argument M cannot vanish when all its coefficients have the same sign.

Theorem 4.1 (Global terminal-prime contraction). *Let M, A arise from a hypothetical nonintegral solution, so in particular $M \geq 2^{14}$ and $q = M/A$. There exists $n_0 = n_0(M, A)$ such that the following holds for every $n \geq n_0$. If $P \in \mathbb{Z}[T] \setminus \{0\}$ satisfies $P(M) = 0$, then*

$$\mathcal{F}_{P,n} \notin \mathbb{Z}.$$

The conclusion is uniform in the degree, span, coefficient height, and sign pattern of P . More precisely, for some negative coefficient index k every prime in I_{n+k} occurs in the reduced denominator of $\mathcal{F}_{P,n}$, and therefore

$$\log \text{den}(\mathcal{F}_{P,n}) \geq (1 - o(1))M^{n+k} \geq (1 - o(1))M^n. \quad (5)$$

The $o(1)$ is along $n + k \rightarrow \infty$.

Proof. Assume for contradiction that $\mathcal{F}_{P,n}$ is an integer. Fix an index k with $c_k < 0$ and choose a prime $p \in I_{n+k}$, which exists for all sufficiently large n by (2). Put $t = n + k$. By Theorem 3.1, p occurs once in F_t , occurs in no lower factor, and satisfies the displayed upper bound in every higher factor. Higher negative coefficients only make the valuation smaller, so integrality implies

$$\begin{aligned} c_k^- &\leq \sum_{j>k} c_j^+ V_p(n+j) \\ &\leq 2q^{n+k} \sum_{j>k} c_j^+ M^{j-k} + 2 \sum_{j>k} c_j^+ \left(1 + \frac{j-k}{n+k}\right). \end{aligned} \quad (6)$$

Multiply by M^k and sum (6) over all negative indices k . For the transported-multiple term, if $T_k = \sum_{j>k} c_j^+ M^j \leq S_+(P)$, then

$$2 \sum_{c_k < 0} q^{n+k} T_k \leq 2q^n S_+(P) \sum_{k \geq 0} q^k = \frac{2q^n}{1-q} S_+(P). \quad (7)$$

For the prime-power allowance, interchange the order of summation. Since $n + k \geq n$,

$$\begin{aligned} &2 \sum_{c_k < 0} \sum_{j>k} c_j^+ M^k \left(1 + \frac{j-k}{n+k}\right) \\ &\leq 2 \sum_{c_j > 0} c_j M^j \sum_{\ell \geq 1} M^{-\ell} \left(1 + \frac{\ell}{n}\right) \\ &= 2S_+(P) \left(\frac{1}{M-1} + \frac{M}{n(M-1)^2}\right). \end{aligned} \quad (8)$$

Combining (4)–(8) gives

$$1 \leq \kappa_n := \frac{2q^n}{1-q} + \frac{2}{M-1} + \frac{2M}{n(M-1)^2}. \quad (9)$$

But $M \geq 2^{14}$ and $q \leq (2/3)^{14}$, so $\kappa_n < 1$ (indeed $\kappa_1 < 0.0072$ at the extremal values, and it decreases with n). This contradiction proves nonintegrality.

For the quantitative statement, observe that if the right-hand side of (6) dominated c_k^- for every negative k , the preceding summation would again give (9). Hence for some k the strict reverse inequality holds. The bound in Theorem 3.1 is uniform over $p \in I_{n+k}$, so every such prime has negative valuation in $\mathcal{F}_{P,n}$. Summing their logarithms and applying (2) gives (5). $\square$

New result — closure of the factorial target

The theorem needs only $P(M) = 0$, one of the three archimedean annihilations. It therefore rules out, for all sufficiently large n , every polynomial contemplated by the unified report’s linear-width factorial feasibility target. No choice of degree, critical span, coefficient height, additional mode cancellations, or structural-prime congruences can make that particular falling-factorial cocycle integral.

Several features deserve emphasis.

- (i) The proof does not pick the highest negative coefficient. It uses the terminal window for every negative coefficient and then invokes the exact global balance $P(M) = 0$.
- (ii) There is no factor equal to the number of negative coefficients: the geometric weights q^k sum to $(1 - q)^{-1}$. This is why the theorem remains uniform even when the degree is very large.
- (iii) The additive “one per prime power” error is not accumulated as a degree factor either; after M -weighting its kernel is $M^{-\ell}$, whose first two moments are explicit.
- (iv) The finite exclusion $M \geq 2^{14}$ is far more than is needed. Asymptotically the contraction requires essentially $2/(M - 1) < 1$, together with sufficiently small q^n .

Corollary 4.2 (No integral near-unit factorial cocycles). *There is no sequence $P_n \in \mathbb{Z}[T] \setminus \{0\}$ such that, for all sufficiently large n ,*

$$P_n(M) = P'_n(M) = P_n(M^2/A) = 0, \quad \mathcal{F}_{P_n,n} \in \mathbb{Z}, \quad \mathcal{F}_{P_n,n} \longrightarrow 1.$$

The conclusion remains true after imposing any additional exact mode cancellations.

Proof. The first displayed equality is already enough for [Theorem 4.1](#). □

4.1 Binomial and occupancy-normalized variants

Put

$$B_n = \binom{A^n}{M^n} = \frac{F_n}{M^n!}, \quad R_n = \frac{F_n}{(A^n)M^n}.$$

For $p \in I_t$ one has $p > M^t$, hence $v_p(M^t!) = 0$ and $v_p(B_t) = v_p(R_t) = 1$. At a higher index,

$$v_p(B_s) \leq V_p(s), \quad v_p(R_s) = V_p(s),$$

because the normalizing factors introduce no new positive high-prime valuation. The same contraction proof therefore gives the following.

Corollary 4.3 (Binomial and no-collision closure). *For all sufficiently large n and every nonzero $P \in \mathbb{Z}[T]$ with $P(M) = 0$, neither*

$$\prod_j B_{n+j}^{c_j} \quad \text{nor} \quad \prod_j R_{n+j}^{c_j}$$

is an integer. In particular, the binomial-cocycle variant of the linear-width target is also infeasible.

5 A critical-window sharpening

The global contraction supersedes the span analysis for the factorial target. Nevertheless, a sharper form of the earlier prime-window theorem is independently useful: the fixed linear margin εn can be replaced by any additive margin tending to infinity. This identifies the genuinely critical boundary before the global mass balance is used.

Let k_n be the highest negative coefficient index of P_n , let $d_n = \deg P_n$, and put

$$t_n = n + k_n, \quad D_n = d_n - k_n, \quad c_* = \log_2 \frac{3}{2} = \log_2 3 - 1 = 0.5849625007 \dots$$

Also set

$$\delta_0 = 2 - \log_2 3 = 0.4150374993 \dots, \quad \Delta_0 = 1 - \frac{8q}{\delta_0(1-q)}.$$

For $x \geq 14$, $\Delta_0 \geq 0.9337$.

Theorem 5.1 (Additive critical-window obstruction). *Let $P_n \in \mathbb{Z}[T]$ be nonzero and have a negative coefficient. Suppose $t_n \rightarrow \infty$ and there is a sequence $h_n \rightarrow \infty$ such that*

$$D_n \leq c_* t_n - h_n. \tag{10}$$

Then

$$\log \text{den}(\mathcal{F}_{P_n, n}) \geq (-c_{n, k_n})(\Delta_0 - o(1))M^{t_n}. \tag{11}$$

In particular, an integral factorial cocycle must satisfy $D_n \geq c_ t_n - O(1)$.*

Proof. The proof follows the prime-window argument imported from the unified report; only the boundary bookkeeping changes. Put $X = A^{t_n}$ and $H = M^{t_n}$. A prime in $(X - H, X]$ is lost from the denominator only if some higher positive factor captures a transported multiple. At relative shift ℓ , the relevant prime interval has length $O(Hq^\ell)$, and the number of possible multipliers is $O(1 + A^\ell H/X)$.

The minimum logarithmic thickness occurs at $\ell = D_n$. Using $c_* = -\log q / \log M$ and (10),

$$\begin{aligned} \frac{\log(Hq^{D_n})}{\log X} &\geq \frac{t_n \log M + (c_* t_n - h_n) \log q}{t_n \log A} \\ &= \delta_0 + \frac{h_n(\log A - \log M)}{t_n \log A} \geq \delta_0. \end{aligned}$$

Thus Brun–Titchmarsh may be applied with any fixed $\delta < \delta_0$ for all large n . The first multiplier sum covers at most

$$\left(\frac{8q}{\delta(1-q)} + o(1) \right) H$$

of logarithmic prime mass. The drifting-multiplier sum is smaller by the factor

$$q^{t_n} M^{D_n} \leq M^{-h_n} \longrightarrow 0,$$

which is exactly where a divergent additive margin replaces the earlier fixed εt_n . Huxley’s theorem supplies total mass $(1 + o(1))H$. Letting $\delta \uparrow \delta_0$ proves (11). $\square$

Remark 5.2. The theorem leaves only a bounded additive strip $D_n = c_* t_n + O(1)$. Canonical Hermite stencils initially appear capable of living in this strip. The next sections show why coefficient multiplicity, and ultimately the global contraction, destroys that appearance.

6 Coefficient-sensitive prime-window duality

The span theorem records whether a higher factor can capture a terminal prime at least once. It ignores how many copies are needed. If the negative coefficient is much larger than the positive coefficient attached to the transported interval, a captured prime can still remain in the denominator. [Theorem 3.1](#) yields an explicit coefficient-sensitive dual certificate.

For a polynomial $P = \sum c_j T^j$ and a negative index k , define

$$\Psi_{n,k}(P) := c_k^- - \sum_{j>k} c_j^+ \left[2q^{n+k} M^{j-k} + 2 \left(1 + \frac{j-k}{n+k} \right) \right]. \quad (12)$$

Proposition 6.1 (Window-dominance certificate). *If $\Psi_{n,k}(P) > 0$ and $n+k$ is sufficiently large, then every prime in I_{n+k} has negative valuation in $\mathcal{F}_{P,n}$. Consequently*

$$\log \text{den}(\mathcal{F}_{P,n}) \geq (1 - o(1))M^{n+k}.$$

Proof. For $p \in I_{n+k}$, lower factors are absent, the factor at k contributes $-c_k^-$, higher negative coefficients may be discarded in an upper bound, and higher positive coefficients are bounded by [Theorem 3.1](#). Thus $v_p(\mathcal{F}_{P,n}) \leq -\Psi_{n,k}(P) < 0$. Sum $\log p$ and use Huxley. $\square$

The proof of [Theorem 4.1](#) can now be read as a Farkas-type statement: if all window-dominance certificates failed, their M^k -weighted sum would imply $S_-(P) \leq \kappa_n S_+(P)$, contradicting $P(M) = 0$. Hence at least one of the explicit rows (12) separates the coefficient vector from the valuation cone.

Corollary 6.2 (Necessary tail inflation without mass balance). *If $\mathcal{F}_{P,n}$ is an integer and $c_k < 0$, then*

$$c_k^- \leq 2q^{n+k} \sum_{j>k} c_j^+ M^{j-k} + 2 \sum_{j>k} c_j^+ \left(1 + \frac{j-k}{n+k} \right).$$

Unless the prime-power allowance dominates, the M -weighted positive tail above k must exceed c_k^- by a factor of order $q^{-(n+k)}$.

This coefficient inflation is the feature missed by a support-only prime-window count. It is also the right diagnostic for exact LP computations: a candidate ray should be rejected before any prime enumeration if one of the inexpensive inequalities (12) is positive.

7 Confluent Hermite tails and complete homogeneous functions

The canonical way to cancel a prescribed finite list of factorial modes while keeping a single high-degree monomial is Hermite interpolation. This section makes that construction exact and shows how the terminal-prime certificate sees it.

Let $\mathcal{R} = (r_1, \dots, r_d)$ be a multiset of positive rational numbers, with repeated entries interpreted confluent. In the applications it contains M twice and begins with

$$(M, M, \lambda), \quad (M, M, \lambda, \eta), \quad (M, M, \lambda, \eta, q), \quad \dots$$

For a polynomial G , let $H_{\mathcal{R}}G$ be the unique polynomial of degree $< d$ that matches G and the required derivatives at the repeated nodes. Thus $G - H_{\mathcal{R}}G$ is divisible over $\mathbb{Q}[T]$ by

$$D_{\mathcal{R}}(T) = \prod_{i=1}^d (T - r_i).$$

Let $h_m(y_1, \dots, y_s)$ denote the complete homogeneous symmetric polynomial of degree m , with $h_0 = 1$.

Lemma 7.1 (Hermite boundary coefficient). *For $e \geq d - 1$, the coefficient of T^{d-1} in $H_{\mathcal{R}}(T^e)$ is*

$$[r_1, \dots, r_d]T^e = h_{e-d+1}(r_1, \dots, r_d).$$

If $\mathcal{R}$ contains M twice, then for every $\ell \geq 0$,

$$h_{\ell}(\mathcal{R}) \geq (\ell + 1)M^{\ell}. \quad (13)$$

Proof. The divided-difference identity follows, for example, by comparing coefficients in

$$\prod_{i=1}^d \frac{1}{1 - r_i z} = \sum_{m \geq 0} h_m(r_1, \dots, r_d) z^m$$

or by induction in the Newton interpolation formula; confluence follows by continuity. For (13), retain only the $\ell + 1$ monomials obtained by distributing total degree ℓ between the two copies of M and assigning degree zero to all other variables. $\square$

A related exact evaluation formula is useful.

Proposition 7.2 (Hermite remainder identity). *For $N \geq d$ let*

$$P_{\mathcal{R},N}(T) = T^N - H_{\mathcal{R}}(T^N).$$

Then, for every z ,

$$P_{\mathcal{R},N}(z) = D_{\mathcal{R}}(z)h_{N-d}(z, r_1, \dots, r_d). \quad (14)$$

Moreover the coefficient of T^{d-1} is $-h_{N-d+1}(\mathcal{R})$.

Proof. Polynomial division of T^N by the monic polynomial $D_{\mathcal{R}}$ gives a quotient whose generating function is the complete homogeneous series. Equivalently, both sides of (14) are monic degree- N polynomials with the same confluent zeros and the same quotient recurrence. The boundary coefficient is [Theorem 7.1](#). $\square$

Because $P_{\mathcal{R},N}$ has d positive roots counted with multiplicity and at most $d + 1$ nonzero coefficients (the degrees $0, \dots, d - 1, N$), Descartes' rule forces every low coefficient to be nonzero and the signs to alternate. In particular the highest low coefficient, at degree $d - 1$, is negative.

The construction extends from one high monomial to an arbitrary nonnegative high tail.

Definition 7.3 (Positive-tail Hermite annihilator). Let

$$G(T) = \sum_{e \geq d} a_e T^e, \quad a_e \in \mathbb{Q}_{\geq 0}, \quad G \neq 0,$$

with finite support, and put

$$P(T) = G(T) - H_{\mathcal{R}}G(T).$$

After multiplying by a positive integer, suppose $P \in \mathbb{Z}[T]$. Then P is called a positive-tail Hermite annihilator for $\mathcal{R}$.

Its coefficient at T^{d-1} is exactly

$$c_{d-1} = - \sum_{e \geq d} a_e h_{e-d+1}(\mathcal{R}), \quad (15)$$

whereas all coefficients above degree $d - 1$ are the nonnegative a_e .

8 Terminal-prime multiplicity mismatch for Hermite tails

The next theorem is subsumed by the global contraction, but it exposes the exact local mechanism and remains useful for symbolic dual certificates.

Theorem 8.1 (Positive-tail Hermite no-go). *Let $\mathcal{R}$ be any finite multiset of positive rational nodes containing M twice, and let $P \in \mathbb{Z}[T]$ be a nonzero positive-tail Hermite annihilator for $\mathcal{R}$. Then for all sufficiently large n , $\mathcal{F}_{P,n}$ is not an integer. More precisely, every prime in I_{n+d-1} occurs in its denominator.*

Proof. Put $t = n + d - 1$ and write $\ell = e - d + 1 \geq 1$. By (15) and (13), the negative boundary coefficient has magnitude at least

$$c_{d-1}^- \geq \sum_{e \geq d} a_e (\ell + 1) M^\ell.$$

For a prime $p \in I_t$, the only positive contributions come from the high tail, and Theorem 3.1 bounds them by

$$\sum_{e \geq d} a_e \left[2q^t M^\ell + 2 \left(1 + \frac{\ell}{t} \right) \right].$$

Since $M \geq 2^{14}$ and $q^t \rightarrow 0$, for every $\ell \geq 1$ and all large t one has

$$(\ell + 1)M^\ell > 2q^t M^\ell + 2 \left(1 + \frac{\ell}{t} \right).$$

Thus the boundary negative multiplicity cannot be supplied by the transported positive tail, and the valuation is negative for every $p \in I_t$. $\square$

For the one-spike stencil $P_{\mathcal{R},N}$, put $D = N - d + 1$. The ratio of the negative boundary coefficient to the top coefficient is

$$h_D(\mathcal{R}) \geq (D + 1)M^D,$$

while the top factorial contains at most $O(q^t M^D + 1 + D/t)$ copies of a terminal prime. The common factor M^D cancels in the comparison; the remaining ratio is essentially $(D + 1)/q^t$, so increasing the span only makes the multiplicity mismatch worse.

Corollary 8.2 (All isolated-top maximal-contact families are closed). *Choose any finite or varying list of additional positive factorial modes, append it to the confluent root list (M, M, λ) , and form the primitive integer multiple of $T^N - H_{\mathcal{R}}(T^N)$. Its factorial cocycle is nonintegral for all sufficiently large n , regardless of how N , the number of cancelled modes, or the coefficients vary.*

This includes the initially promising families that cancel η , then q , then τ , and so on. The obstruction is stronger than rational-root denominator growth and stronger than the critical-span theorem: it does not require archimedean near-unit estimates at all.

9 The apparent mode phase diagram

It is still instructive to record why the Hermite route looked promising before the multiplicity mismatch was included. Put

$$R = \frac{\log A}{\log M} = \log_2 3 = 1.5849625007 \dots, \quad c_* = R - 1.$$

For a subunit mode $\rho = M^s/A^r$, define its normalized decay deficit

$$\delta(\rho) = \log_M(1/\rho) = rR - s > 0.$$

The first modes from [Theorem 2.1](#) have the following data.

rank	mode	(r, s)	coefficient	δ	numerical value
1	$\eta = M^3/A^2$	(2, 3)	$-1/6$	$2R - 3$	0.1699250
2	$q = M/A$	(1, 1)	$+1/2$	$R - 1$	0.5849625
3	$\tau = M^4/A^3$	(3, 4)	$-1/12$	$3R - 4$	0.7548875
4	$\sigma = M^2/A^2$	(2, 2)	$+1/4$	$2R - 2$	1.1699250
5	$\nu = M^5/A^4$	(4, 5)	$-1/20$	$4R - 5$	1.3398500

Suppose a one-spike Hermite stencil cancels all modes through a subunit root ρ . If $\rho = u/v$ in lowest terms and the low footprint has fixed degree, evaluating the integer stencil at u/v gives the elementary gap divisibility

$$v^D \mid \text{lc}(P),$$

where D is the exponent gap between the isolated top monomial and the low footprint. Since $v \geq 1/\rho$, the primitive leading coefficient costs at least ρ^{-D} . If ρ_- is the last cancelled mode and ρ_+ the first uncanceled mode, the naively available slope is therefore at most

$$\frac{\delta(\rho_+)}{1 + \delta(\rho_-)}. \quad (16)$$

The first values are

last cancelled	first surviving	bound from (16)	comparison with c_*
none	η	0.1699250	below
η	q	1/2	below
q	τ	0.4762810	below
τ	σ	2/3	above
σ	ν	0.6174638	above

Thus the first scalar-height calculation not automatically killed by the prime threshold is the stencil cancelling η, q, τ , whose first surviving mode is σ . This is the “first surviving seven-term stencil” found during the present investigation. It does not actually survive: [Theorem 8.1](#) shows that its highest low negative coefficient requires $(D + 1)M^D$ copies of each terminal prime, while all transported positive factors supply only $O(q^t M^D)$. The global contraction then closes every non-Hermite variation as well, provided $P(M) = 0$ and integrality is demanded.

10 What exactly has been closed

The unified report posed the following feasibility question in substance: can one find varying integer polynomials P_n such that

$$P_n(M) = P'_n(M) = P_n(M^2/A) = 0, \quad v_p(\mathcal{F}_{P_n, n}) \geq 0 \text{ for every } p, \quad \mathcal{F}_{P_n, n} \rightarrow 1,$$

with span at or above the threshold $c_*(n + k_n)$? The answer is now unconditionally negative under the hypothetical candidate assumptions.

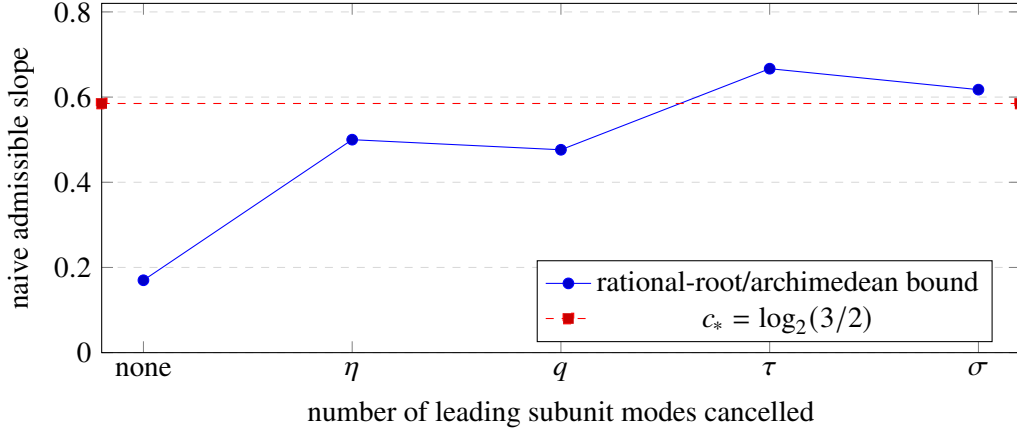


Figure 1: The scalar-height phase diagram falsely suggests an escape after cancelling τ . Terminal-prime multiplicity, absent from this graph, closes every displayed Hermite family.

Theorem 10.1 (Negative resolution of factorial feasibility). *No such sequence P_n exists. The span condition and the near-unit condition are superfluous: for every sufficiently large n , the conjunction*

$$P_n \neq 0, \quad P_n(M) = 0, \quad v_p(\mathcal{F}_{P_n, n}) \geq 0 \text{ for every prime } p$$

is already impossible.

Proof. This is [Theorem 4.1](#). □

The result should not be misread as progress toward a contradiction from the candidate itself. It proves that the proposed auxiliary integers cannot exist. In the proof-search ledger, this is nevertheless valuable for three reasons.

1. It removes a named open direction from the prior report rather than merely exhibiting another failed example.
2. It gives an explicit, low-complexity dual certificate. Future LP/ILP searches need not spend time on the factorial valuation cone once $P(M) = 0$ is imposed.
3. It isolates the structural reason: the same base M controls both the archimedean annihilation and the coefficient mass balance, while the high-prime transport loses the factor q^{n+k} . Any viable replacement must break that contraction, not just improve a constant.

11 How a future construction must evade the contraction

The contraction proof has four essential hypotheses:

- H1.** the auxiliary object is a signed product of terminal intervals indexed by one shift variable;
- H2.** a prime in the base terminal interval occurs with unit multiplicity;
- H3.** higher transport is bounded by $q^t M^\ell$ plus a summable prime-power error;
- H4.** the archimedean cancellation supplies the same-base balance $P(M) = 0$.

A successful strategy derived from this circle must violate at least one of them. Merely moving to the critical span does not help. The following interfaces appear genuinely different.

11.1 Two-dimensional coefficient transport

Instead of a one-variable shift polynomial, use a coefficient array c_{ij} on a two-parameter family whose archimedean cancellations are

$$\sum_{i,j} c_{ij} M^i A^j = 0, \quad \sum_{i,j} (i,j) c_{ij} M^i A^j = 0.$$

A terminal prime could then be transported in two noncomparable directions. The one-variable geometric kernel q^k would be replaced by a genuinely two-dimensional operator. The first question is whether its positive kernel still has norm < 1 after the exact mass balances are inserted. If it does, the present proof generalizes and closes the construction; if it does not, the escaping direction identifies where an external structural prime might enter.

Remaining target — two-dimensional contraction test

Choose a concrete two-parameter terminal product with exact valuation formula and compute the $M^i A^j$ -weighted transport norm. Prove either a contraction theorem analogous to [Theorem 4.1](#), or exhibit an explicit normalized positive eigenvector of norm at least one whose support is incompatible with the integer-exponent controls.

11.2 Structural-prime weights rather than terminal-prime weights

At a private structural prime $r \mid M, r \nmid A$, Kummer’s theorem gives long carry-free prefixes in $\binom{A^n}{M^n}$. The unified report correctly observed that the raw valuation is only $O(n)$, whereas a terminal-prime window has mass M^n . The contraction theorem changes the relevant comparison: critical coefficient vectors would need exponentially inflated tails, so an $O(n)$ structural valuation multiplied by those coefficients may no longer be negligible.

A promising hybrid certificate should combine

$$\sum_{p \in I_t} (\log p) v_p(\cdot) \quad \text{with} \quad \gamma_t v_r(\cdot)$$

for a carefully selected weight γ_t . The terminal part enforces the coefficient inflation; the structural part should then rule out the inflated vector using carry-free prefixes. This is different from adding a linear structural gain to a fixed-coefficient archimedean estimate.

Remaining target — coefficient-inflated Kummer dual

For the binomial valuation vectors $b_j = v_r \left(\binom{A^{n+j}}{M^{n+j}} \right)$, find a nonnegative combination of terminal-window rows and one structural row that is strictly negative on the annihilator subspace $P(M) = P'(M) = P(\lambda) = 0$, without assuming a span bound. Exact rational Farkas certificates for moderate (n, d) should be searched first; the symbolic target is a geometric weight recurrence rather than a dense LP solution.

11.3 Primitive-sensitive normalization

The contraction theorem is satisfied by the controls $(2^m, 3^m)$ and therefore cannot prove the conjecture. A positive route must consume a fact absent for controls. The three clean candidates imported from the unified report remain:

1. an external prime in the support of MA ;

2. the least-generator decomposition of the solution monoid;
3. saturation of the rational-output spectrum to the 3-smooth bases.

A useful design rule follows. Before investing in a new auxiliary construction, write down its analogue for $(M, A) = (2^m, 3^m)$. If every input and every inequality remains valid, the construction can at most close a method negatively. To prove the conjecture, one line of the argument must visibly fail for the controls.

12 A recursive Hermite-peeling program

Although the global contraction closes factorial integrality outright, the algebra behind it suggests a reusable decomposition for other auxiliary sequences.

Let $D_0(T) = (T - M)^2(T - \lambda)$ over $\mathbb{Q}[T]$, and let P be divisible by D_0 . If k is the highest negative coefficient index, split

$$P = G + L, \quad G = \sum_{j>k} c_j T^j \quad (c_j \geq 0), \quad \deg L \leq k.$$

Let $H = H_{(M,M,\lambda)}G$. Since P and $G - H$ are divisible by D_0 ,

$$P = (G - H) + D_0 Q, \quad \deg Q \leq k - 3. \quad (17)$$

The first summand is a positive-tail Hermite atom and has a completely explicit negative boundary coefficient. The second is a strictly lower-degree annihilator. Repeating (17) produces a hierarchy of atoms.

For factorial cocycles, the global contraction makes the hierarchy unnecessary. For a future two-dimensional or primitive-sensitive construction, however, it may provide the right sparse dual basis: terminal windows are triangular in the cutoff index, while Hermite atoms are triangular in degree.

Problem 12.1 (Hermite-peeling duality). *Construct weights on the valuation windows associated with successive cutoffs in (17) so that every positive-tail atom contributes negatively and all cross-atom transport is dominated by a summable triangular kernel. Determine whether an external structural prime can change the spectral radius from < 1 to ≥ 1 only for primitive candidate pairs.*

13 Exact computational protocol

The new theorems are symbolic, but exact computation remains useful for validation and for searching the next hybrid dual. No floating-point factorials are required.

13.1 Generating maximal-contact stencils

For a rational root multiset $\mathcal{R}$ and degree N , compute the remainder of T^N modulo $D_{\mathcal{R}}$ over $\mathbb{Q}[T]$ and clear denominators. In Sage/Python-like notation:

```
# T is a polynomial indeterminate over QQ.
D = prod(T - r for r in roots) # repeat M twice
H = (T**N).mod(D) # confluent remainder is automatic
P = T**N - H
P = primitive_integer_multiple(P) # positive leading coefficient
assert all(P(r) == 0 for r in distinct_roots)
```


For repeated roots one also verifies the derivative condition. The identity (14) gives a second, independent check.

For the control pair $(M, A) = (2, 3)$ and gap $D = 6$, primitive one-spike stencils have the following negative-boundary/top ratios. The universal lower bound is $(D + 1)M^D = 448$.

d	cancelled root multiset beyond (M, M, λ)	degree N	$-c_{d-1}/c_N$
3	none	8	$716224/729 \approx 982.47$
4	η	9	$806893504/531441 \approx 1518.31$
5	η, q	10	$1089702784/531441 \approx 2050.47$
6	η, q, τ	11	$1027112873344/387420489 \approx 2651.16$

The ratios grow, while the transported multiplicity of a terminal prime carries the additional small factor q^t .

13.2 Searching hybrid certificates

For fixed (M, A, n, d) , assemble the exact valuation matrix

$$V_{p,j} = v_p(F_{n+j})$$

only for the following sparse rows:

1. terminal-window primes for every candidate negative cutoff;
2. their transported multiples in higher intervals;
3. one or two structural-prime rows;
4. the three exact annihilator rows M^j , jM^j , and λ^j .

First solve the rational Farkas dual. An exact certificate is a vector of rational row weights whose combination has the desired sign on every coefficient coordinate. Only after a stable sparse pattern appears should one impose integer coefficients or prime-power rows. The contraction proof predicts geometric terminal weights; the remaining unknown is the structural row weight.

14 Lean formalization blueprint

Most of the new argument is elementary and well suited to formalization once the analytic prime-existence input is isolated. A proposed module is `FactorialCocycleContraction`. The dependency structure is:

1. **Finite interval arithmetic.** Formalize F_t , $V_p(t)$, the fact that $p \in (A^t - M^t, A^t]$ gives $V_p(t) = 1$, and absence from lower factors.
2. **Higher valuation bound.** Prove a floor-sum inequality

$$V_p(t + \ell) \leq \frac{M^{t+\ell}}{p-1} + \left\lfloor \frac{(t + \ell) \log A}{\log p} \right\rfloor$$

and then a rational, logarithm-free coarse bound sufficient for the contraction. Since the candidate has $M \geq 16384$ and $A \geq 3^{14}$, very loose integer constants are acceptable.

3. **Sign-mass identity.** For integer coefficients and $P(M) = 0$, prove

$$\sum_j c_j^+ M^j = \sum_j c_j^- M^j.$$

4. **Geometric summation.** Formalize the two finite-sum bounds by comparison with $\sum_{\ell \geq 1} M^{-\ell}$ and $\sum_{\ell \geq 1} \ell M^{-\ell}$.
5. **Abstract prime-window hypothesis.** State the contraction theorem assuming that each sufficiently large interval $(A^t - M^t, A^t]$ contains a prime. The paper-level Huxley theorem then discharges this single hypothesis outside the kernel, while finite ranges can be checked computationally if desired.
6. **Hermite algebra.** The complete-homogeneous identity can be formalized independently using polynomial remainder or divided differences. Only positivity and the $(\ell + 1)M^\ell$ lower bound are needed for the no-go corollary.

The analytic theorem need not be encoded as an opaque global axiom. A clean interface is a structure carrying a function that returns a prime and proofs of its two endpoint inequalities for every $t \geq t_0$. This keeps every arithmetic use kernel-checked and makes the external dependency explicit.

15 Failure checks and robustness

A result this short relative to the size of the prior search deserves aggressive checking. The following possible loopholes were tested in the derivation.

15.1 Could one terminal prime cancel several negative coefficients?

Each negative index uses a different terminal interval I_{n+k} . The proof never assumes the chosen primes are distinct, although the intervals are in fact disjoint for large indices. More importantly, the coefficient inequalities are summed only after multiplying by M^k ; no counting of distinct primes is used in the contraction step.

15.2 Could high prime powers invalidate the transport estimate?

They are exactly the source of the second term in (3). After summing over coefficient indices, their contribution is governed by

$$\sum_{\ell \geq 1} M^{-\ell} \quad \text{and} \quad \sum_{\ell \geq 1} \ell M^{-\ell},$$

not by the degree. This is the reason for retaining the apparently crude “+1 per prime power” term instead of suppressing it in $O(1)$ notation.

15.3 Could extremely many coefficients defeat the proof?

No. The transported-multiple term is weighted by q^{n+k} , and the negative indices are distinct nonnegative integers, so

$$\sum_{c_k < 0} q^k \leq \frac{1}{1 - q}.$$

The degree never enters. The polynomial only needs finite support so that the cocycle is defined.

15.4 Could huge coefficients defeat the proof?

No. Every inequality is homogeneous in the coefficient vector. The exact equality $S_+ = S_-$ and the contraction compare relative mass, not absolute height.

15.5 Does the proof accidentally assume the conjecture?

No. It assumes the standard hypothetical pair (M, A) and the already verified lower bound $M \geq 2^{14}$. The same conclusion holds for the control pairs, as a negative method-closure must. It does not infer $x \in \mathbb{Z}$.

15.6 Is Huxley used uniformly in a moving coefficient index?

The only requirement is that every interval I_t contain a prime once t exceeds a fixed threshold. Huxley’s asymptotic supplies precisely such an eventual statement. Since $t = n + k \geq n$, choosing n beyond that threshold handles every coefficient index simultaneously, even when the degree depends on n .

16 Consequences for the global research program

The attached unified report ended with four explicit directions: negative closure of the linear-width factorial family, shared-face masks, a quadratic-gain structural mechanism, and further finite-certificate doublings. The first is now closed, and the closure is stronger than anticipated.

Prior direction	Updated status
Linear-width factorial feasibility	Closed negatively by Theorems 4.1 and 10.1 ; degree and span are irrelevant.
Shared-face masks	Still open; the contraction suggests computing a two-dimensional transport norm before a full integer program.
Quadratic-gain structural mechanism	Still open, but coefficient inflation changes the correct scale for a hybrid terminal/structural dual.
Finite-certificate doublings	Orthogonal and still useful; they strengthen the numerical lower bound on a hypothetical exponent.

The most important conceptual update is that the critical constant $\log_2(3/2)$ was not the true obstacle for factorial cocycles. It arose when one asked only whether a terminal prime could be transported once. The annihilation $P(M) = 0$ imposes a coefficient mass balance, and after that balance is summed over all negative windows the transport operator is a strict contraction. The critical-width LP therefore had no hidden feasible ray waiting for a structural-prime refinement.

Caution — what this does not establish

The Alaoglu–Erdős conjecture remains unproved in this report. The new theorem says that a particular proposed source of auxiliary integers cannot work. It neither supplies a counterexample nor upgrades Four Exponentials technology. Its value is a rigorous removal of a large search region and a sharper specification of what a successful primitive-sensitive construction must do differently.

17 Next concrete experiments

The following sequence is the most efficient continuation.

1. **Formalize the contraction core.** This is small compared with the existing formal corpus and would turn the main new theorem into a kernel-checked conditional on a single prime-window interface.
2. **Build the two-dimensional transport matrix.** Start with terminal products indexed by a rectangular (i, j) grid and compute the norm after both exact mass balances. Do this first on controls and then on synthetic primitive-support pairs.
3. **Add one structural row.** Search exact Farkas certificates, not approximate floating-point LP solutions. Inspect whether the row weight is stable under increasing n, d .
4. **Test control sensitivity at every stage.** Discard any purported positive argument that remains valid word-for-word for $(2^m, 3^m)$.
5. **Continue the finite Lean frontier independently.** It improves all numerical constants, but should not be conflated with the structural proof search.

18 Conclusion

The main advance is a global use of terminal-prime windows. For each negative coefficient, integrality demands upward transport of its terminal primes. After weighting by M^k , the transported-multiple kernel contributes at most $2q^n/(1-q)$ of the total positive mass, and the prime-power kernel contributes at most $2/(M-1) + 2M/(n(M-1)^2)$. The exact annihilation $P(M) = 0$ says that positive and negative masses are equal. For a hypothetical candidate, the resulting operator norm is far below one. This contradiction is uniform over the entire coefficient space.

Accordingly, the factorial and binomial linear-width programs are not merely difficult at the critical span; they are infeasible at every span. The Hermite analysis explains the local multiplicity mismatch, and the additive critical-window theorem sharpens the earlier support-only obstruction, but the mass-balance contraction is the decisive closure.

The conjecture itself now has one fewer plausible hiding place. The next serious attempt should be two-dimensional and primitive-sensitive from the outset, with the transport norm computed before any elaborate interpolation or formalization is undertaken.

A Exact constants

For reference,

$$\begin{aligned}
 \alpha &= \frac{\log 2}{\log 3} = 0.6309297535714574\dots, \\
 R &= \alpha^{-1} = \log_2 3 = 1.5849625007211562\dots, \\
 c_* &= R - 1 = 0.5849625007211562\dots, \\
 \delta_0 &= 2 - R = 0.4150374992788438\dots, \\
 q_{\max} &= (2/3)^{14} = 0.0034254873907817\dots, \\
 \Delta_0(q_{\max}) &= 1 - \frac{8q_{\max}}{\delta_0(1 - q_{\max})} = 0.9337455204\dots
 \end{aligned}$$

At the extremal candidate bounds $M = 2^{14}$, $q = q_{\max}$, the contraction constant is

$$\kappa_1 = 0.0071186864\dots, \quad \kappa_{10} = 0.0001342863\dots$$

The asymptotic limit of the displayed coarse bound is $2/(2^{14} - 1) = 0.0001220778\dots$

B The rational-root exponent-gap lemma

Lemma B.1 (Exponent-gap divisibility). *Let*

$$Q(T) = aT^N + \sum_{j=0}^b c_j T^j \in \mathbb{Z}[T], \quad a \neq 0, \quad b < N,$$

and suppose u/v in lowest terms is a nonzero rational root. Then

$$v^{N-b} \mid a.$$

Proof. Multiply $Q(u/v) = 0$ by v^N :

$$au^N + \sum_{j=0}^b c_j u^j v^{N-j} = 0.$$

Every term in the sum is divisible by v^{N-b} , so $v^{N-b} \mid au^N$. Since $\gcd(u, v) = 1$, the same power divides a . $\square$

For a one-spike Hermite stencil, $b = d - 1$ and $N - b = D$. If the last cancelled subunit mode is $\rho = u/v$, then $v \geq 1/\rho$, producing the scalar slope estimate (16). This lemma is useful diagnostically but is not needed by the global contraction theorem.

C General factorial mode enumeration

Faulhaber’s formula gives an algorithmic expansion

$$\log F_n = nM^n \log A - \sum_{r \geq 1} \frac{1}{r} A^{-rn} \sum_{j=0}^{M^n-1} j^r.$$

If

$$\sum_{j=0}^{K-1} j^r = \sum_{s=1}^{r+1} \gamma_{r,s} K^s,$$

then every nonzero coefficient contributes the mode

$$-\frac{\gamma_{r,s}}{r} \left(\frac{M^s}{A^r} \right)^n.$$

The subunit modes are ordered by the positive deficits $rR - s$. The first twelve are:

rank	r	s	deficit $rR - s$	coefficient
1	2	3	0.1699250014	$-1/6$
2	1	1	0.5849625007	$+1/2$
3	3	4	0.7548875022	$-1/12$
4	2	2	1.1699250014	$+1/4$
5	4	5	1.3398500029	$-1/20$
6	3	3	1.7548875022	$+1/6$
7	5	6	1.9248125036	$-1/30$
8	2	1	2.1699250014	$-1/12$
9	4	4	2.3398500029	$+1/8$
10	6	7	2.5097750043	$-1/42$
11	3	2	2.7548875022	$-1/12$
12	5	5	2.9248125036	$+1/10$

The ordering is a small-denominator Diophantine approximation problem for $R = \log_2 3$. Cancelling a long initial segment by exact rational roots rapidly increases coefficient denominators, but [Theorem 4.1](#) shows that no amount of such cancellation can restore factorial integrality.

D Bibliographic note

The conjecture is a special case of the Four Exponentials problem. The recent matrix coefficient formulation of Dasgupta and Kakde provides a modern structural setting but does not currently prove the 2×2 case. The analytic input used in this report is classical: Huxley’s short-interval prime theorem and Brun–Titchmarsh. No claimed unpublished result by another AI system is used as a mathematical premise.

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